

Video 1 – The Learning Problem

Video Contents

- *Introduction* 00:00 – 02:36
- *Netflix* 02:36 – 13:00
- *Components of Learning* 13:00 – 21:00
- *Solution Components* 21:00 – 23:18
- *The perceptron model* 23:18 – 36:03
- *Supervised, Unsupervised and Reinforcement learning* 36:03 – 49:17
- *A learning puzzle* 49:17 – 55:30
- *SAFE SKIP! Q&A* 55:30 – end

Video Synopsis

Introduction

00:00 – 02:36

Prof. Mostafa introduces the course and explains the structure which revolves around the following topics in that order: 1) What is learning? 2) Can we learn? 3) How can we learn? and 4) How can we learn better? The start of this 'story' will be quite theoretical, but once the theoretical and conceptual aspects have been explored, the course will become more practical.

Netflix

02:36 -13:00

Prof. Mostafa provides an example of use of machine learning by Netflix, which boils down to describing each and every movie/show in terms of thousands of tags. As for each item there are also many ratings from viewers, the job of the machine learner is now to understand the relationship between tags and ratings, such that for a movie/show not yet seen by a user, the algorithm can predict whether the viewer will like the show or not based on its tagged content.

Components of Learning

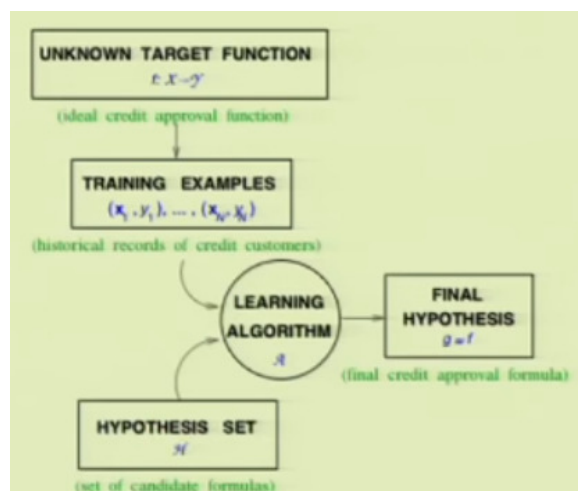
13:00-21:00

Prof. Mostafa introduces some core concepts that together form the components of learning: We will always want to approximate some sort of relationship between input data X and output data Y . The ideal relationship (or the 'real' relationship if you like), is called the (unknown) target function and will always be indicated with f . Mathematically speaking, the target function defines a function from the input domain X , to the output domain Y : $f: X \rightarrow Y$. It is now the job of the machine learner to choose a suitable function from a set of candidate functions, called hypotheses. When focussing on supervised learning (and more about that later) which is the focus of this module, data will always be represented in pairs of input and output data points:

$(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$. It is extremely unlikely that we will ever find the target function, f , exactly.

What we aim for is a final hypothesis $g: X \rightarrow Y$ that approximates the target function reasonably well.

The components of learning can be visualised in this diagram, which will be built upon throughout the course. This diagram shows that the unknown target function is responsible for the creation of data points that we can use for training of our learning model and we hope that this will result in a good final hypothesis. The learning model consists of two elements: the learning algorithm and the hypothesis set. In future lectures it will become clear that choosing a hypothesis set is crucial to proving that learning from data is possible and that we do not constrain ourselves by choosing a hypothesis set.



Solution Components

21:00-23:18

The solution components, or the components in the diagram that we can manipulate, are the hypothesis set and the learning algorithm. An example would be neural networks (as the hypothesis set) and back propagation (as the learning algorithm). Together these are called the learning model.

The perceptron model

23:18 – 36:03

As a simple example of a learning model, the perceptron model is introduced as a linear model that can be used to classify linearly separable classes. Linearly separable means that we can draw a straight boundary line between data points of the various (in this case two) classes with all data points for one class on one side and all data points for the other class on the other side. The perceptron learning algorithm is introduced as an algorithm that, starting from any arbitrarily chosen line, will converge to a line that acts as such a decision boundary as long as the data set is linearly separable. This is quite amazing if you keep in mind that the perceptron algorithm will only ever take one data point in account when updating its candidate boundary line!

Mathematically, a data point is classified as belonging to one class as the vector dot product of the data point with the vector that describes the line results in a negative number and as a point in the other class if this dot product is positive. The perceptron learning algorithm updates the vector (or weights) that describe the boundary line in such a way that the weights are nudged a little bit into the direction that would make a misclassified points 'less' misclassified (i.e. closer to the boundary line). It is very useful to understand the mathematics presented, as the perceptron model gives rise to artificial neural networks.

Supervised, Unsupervised and Reinforcement learning

36:03-49:17

These three types of learning are briefly introduced:

Supervised learning will always use input and output data to determine a hypothesis that best matches this data. So far the lecture has focussed on such methods.

Unsupervised learning is more of an exploratory method in which differences between groups of data points are identified. It is then hoped that these differences provide some sort of information on the meaning of a data point in one such grouping versus a data point in another grouping.

Reinforcement learning is introduced as a completely different type of learning in which the machine learner learns from the inputs to the system, **any** (not necessarily desired) output, and a grade for that output (how beneficial was the output) what the best output for a given input is.

A learning puzzle

49:17-55:30

A learning puzzle is introduced to summarize what learning is, and that it is not trivial that learning is possible. First of all, learning is stated as being able to accurately predict what the value of the function f is for the unclassified example based on learning from the classified examples. Prof.

Mostafa then shows that for this particular example learning is not possible. Based on the given examples, the unknown point could be either -1 or +1! Clearly, we need to know more to determine whether learning from data is possible..

SAFE SKIP! Q&A

55:30 – end